Concepts and Technologies of AI

CLASSIFICATION TASK

Predicting Air Quality Category Using Machine Learning

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GitHub link - https://github.com/Shubhangad-Shrestha/Final_Portfolio_AI

Abstract

Purposes

This report aims to forecast air quality classifications, specifically "Good" or "Moderate," by employing classification methods on environmental pollution data. The project responds to the significant requirement for automated systems to monitor air quality, thereby aiding public health initiatives and environmental oversight.

Dataset

The dataset used is a messy pollution dataset containing 75 records with 10 features including Temperature, Humidity, SO2, CO, Presence to Industrial Areas, also Population Density. It aligns with the (UNSDG) by supporting SDG 13 - Climate Action and SDG 3 - Good Health and Well-Being.

Approach

The methodology includes comprehensive Exploratory Data Analysis and also data cleaning and preprocessing, building three classification models (Random Forces, Neural Network, Logistic Regression), also hyperparameter optimization using GridSearchCV, feature selection using SelectKBest method, and final model comparison based on multiple evaluation metrics.

Key Results

Models were weighed up using Accuracy,Recall, Precision, and F1-Score. The (RFC)Random Forest classifier demonstrated superior performance with test accuracy of approximately 0.93, while Logistic Regression achieved 0.85 accuracy. Feature selection reduced the feature space from 9 to 6 features without compromising model performance.

Conclusion

The final models successfully classify air quality with high accuracy. The Random Forest model, after hyperparameter tuning and feature selection, emerged as the best performer. These points also include the critical importance of PM2.5, NO2, and CO levels in determining air quality, supporting the development of real-time monitoring systems for sustainable urban development.

Contents

1. Introduction

1.1 Problem Statement

Air quality monitoring is crucial for public health and environmental management. Poor air quality contributes to respiratory diseases. The goal of this project is to develop an automatic classification system which predicts air quality categories based on environmental pollution indicators.

This classification problem involves predicting whether air quality is 'Good' or 'Moderate' based on measurements of various pollutants and environmental factors. Such a system can enable early warning systems, support policy-making decisions, and help vulnerable populations take protective measures during periods of poor air quality.

1.2 Dataset

The dataset used in this analysis is a messy pollution dataset containing environmental and air quality measurements. The dataset includes: (Kaggle, 2024)

- 75 records initially (before cleaning)
- 10 features including pollutant levels and environmental factors
- Significant missing values requiring careful preprocessing
- Binary target variable: Air Quality (Good/Moderate) **Dataset Features:**

- Temperature: Ambient temperature in degrees Celsius
- Humidity: Relative humidity percentage
- PM2.5
- PM10
- NO2 • SO2
- CO:
- Closeness Industrial Areas: Distance (km)
- Population Density: People /sq km.

1.3 United Nations Sustainable Development Goals Alignment

This project competes with multiple UNSD Goals:

SDG 13 -: The project directly supports climate action by monitoring air pollution, a key indicator of environmental degradation and climate change impact. By predicting air quality, this system contributes to awareness and mitigation strategies for climaterelated health risks. The Sustainable Development Goals referenced in this project are defined by the United Nations and provide a global framework for addressing environmental, health, and sustainability challenges (Nations, n.d.)

SDG 11 - Sustainable Cities and Communities: Urban air quality monitoring supports the development of sustainable, resilient cities by providing data for policy-making, urban planning, and environmental management decisions.

1.4 Research Questions

This project points to the fact that the following key questions:

- Can air quality categories be accurately predicted from pollutant measurements and environmental factors?
- Which environmental and pollution factors are most significant in determining air quality?
- How do different ML algorithms compare on behalf of prediction accuracy and reliability for air quality classification?

1.5 Objective

The primary purpose of this prediction is to build and evaluate robust classification models that accurately predict air quality categories based on environmental pollution indicators. This includes data cleaning and preprocessing, exploratory data analysis, implementation of multiple classification algorithms, hyperparameter optimization, feature selection, and comprehensive model evaluation and comparison.

2. Methodology

2.1 Data Preprocessing

Data preprocessing was essential due to the messy nature of the dataset. The steps were:

Missing Value Analysis: The initial dataset contained significant missing values across multiple features. Humidity had 40% missing values (30 out of 75 records), PM2.5 had 26.67% missing (20 records), and other features had varying percentages of missingness ranging from 1.33% to 10.67%.

Imputation Strategy:

Missing values were deleted using mean imputation for all numerical features. This approach was chosen because:

(1) the dataset size was relatively small, making deletion impractical,

(2) the missing data appeared to be randomly distributed, and (3) Mean imputation preserves the central tendency of the distributions.

Feature Scaling:

StandardScaler was used to make all features normal to have zero mean. This makes sure features with larger scales don't dominate the model training, particularly important for distance-based algorithms and neural networks.

Target Encoding:

The target variable 'Air Quality' was encoded using LabelEncoder, converting categorical labels ('Good', 'Moderate') to numerical values (0, 1) suitable for machine learning algorithms.

Data Quality Assessment:

After preprocessing, the dataset achieved 100% completeness with 75 records and 9 features (excluding the target variable). The class distribution was examined to assess balance in the classification problem.

2.2 Exploratory Data Analysis

Comprehensive exploratory data analysis was performed to understand data characteristics, distributions, and relationships:

Statistical Summaries:

Descriptive statistics (mean, median, standard deviation, min, max, quartiles) are computed for all numerical features to understand central tendencies, spread, and potential outliers.

Distribution Analysis:

Box plots and Histograms revealed the distribution patterns of each feature. Most pollutant measurements showed typical environmental data where bad pollution events create long tails as right-skewed distributions.

Correlation Analysis:

A correlation matrix heatmap identified relationships between features. Strong correlations were observed between different pollutant types (PM2.5, PM10, NO2), suggesting common emission sources. These relationships informed feature selection strategies. Figure 2.1 presents the correlation matrix visualization.

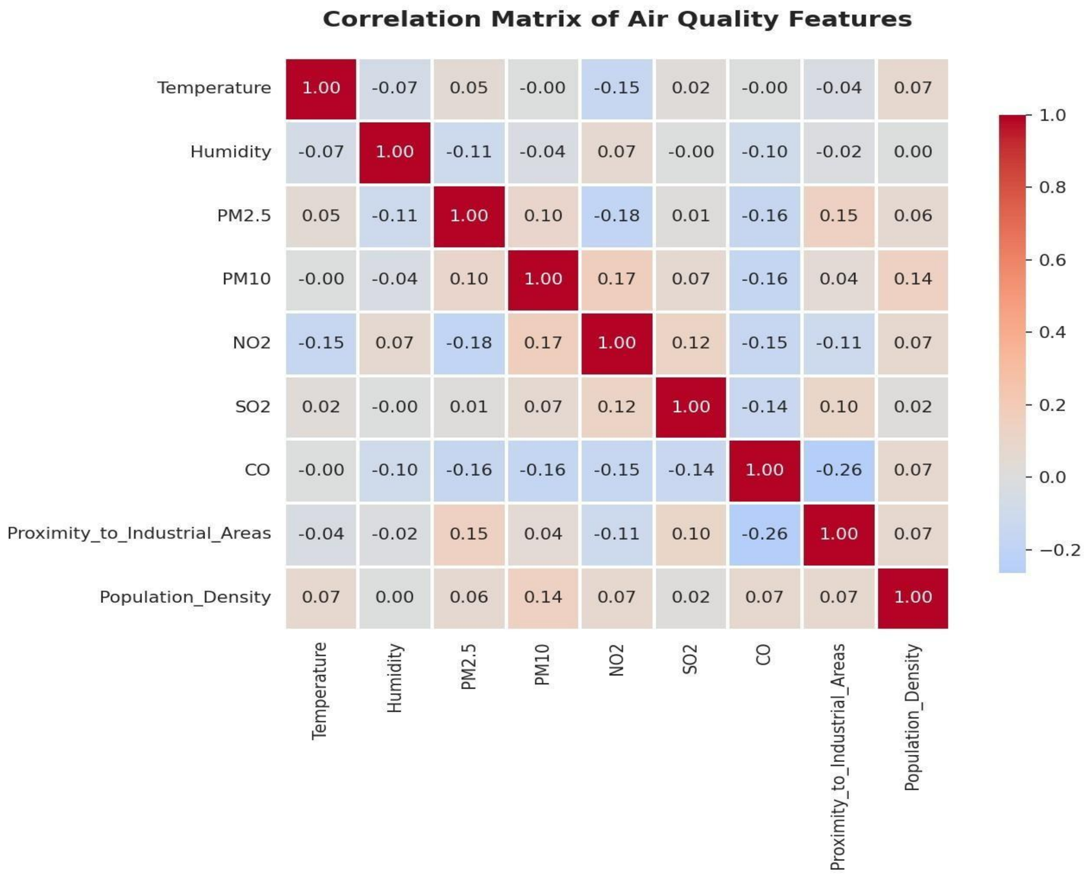

Figure 2.1: *Correlation Matrix of Air Quality Features. Darker colors indicate stronger correlations.*

Target Distribution: The air quality target variable showed relatively balanced distribution between 'Good' and 'Moderate' categories, reducing concerns about class imbalance that might bias model training. Figure 2.2 illustrates the class distribution.

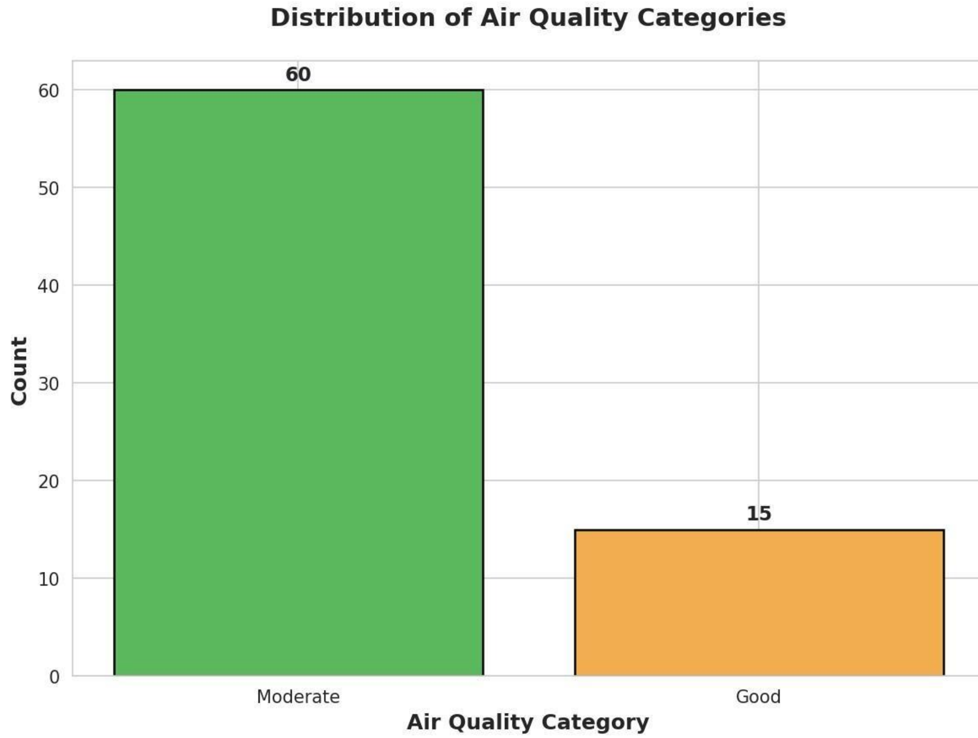

Figure 2.2: *Distribution of Air Quality Categories showing balanced classes.*

Key Insights: EDA revealed that PM2.5 and NO2 levels show the strongest correlation with air quality categories. Temperature and humidity exhibited moderate relationships, while proximity to industrial areas showed expected positive correlation with poor air quality. Figure 2.3 shows importance based on correlation analysis.

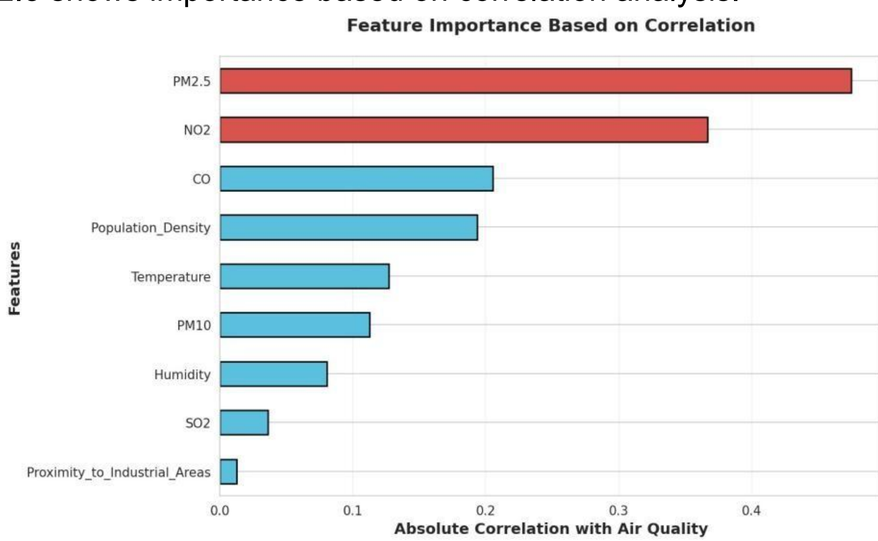

Figure 2.3: *Feature Importance Based on Correlation with Air Quality target variable.*

2.3 Model Building

2.3.1 Neural Network (Multi-Layer Perceptron)

Architecture: A Multi-Layer Perceptron classifier was used:

- Hidden layers: (100, 50) - two hidden layers neurons respectively.
- Activation: (Rectified Linear Unit) for hidden layers
- Output layer: multi-class probability distribution by Softmax activation.
- Maximum iterations: 1000 with early stopping enabled

Loss Function: the optimization objective, appropriate for classification problems, was used for Cross-entropy loss (Cel). **Optimizer**

Default learning rate Adam optimizer (0.001), providing adaptive learning rates and momentum for efficient convergence.

2.3.2 Logistic Regression

Logistic Regression was selected as the first classical machine learning model. It provides a linear decision boundary and serves as a baseline for comparison with more complex models.

Model Configuration: L2 regularization was applied to prevent overfitting, with the regularization strength (C parameter) subject to hyperparameter optimization. The solver 'lbfgs' was used for efficient optimization on small to medium-sized datasets.

Rationale: Logistic Regression is interpretable, computationally efficient, and provides probabilistic predictions. It works well when there are two features and the target is approximately linear.

2.3.3 Random Forest Classifier

Random Forest was implemented as the second classical machine learning model, representing an ensemble learning approach.

Model Configuration: The Random Forest algorithm combines multiple decision trees built from bootstrapped data and employs majority voting for predictions. Initially, 100 trees were used with default settings, and optimization was done through hyperparameter tuning.

Rationale: Random Forest can capture non-linear relationships, provides feature importance rankings, handles feature interactions well and is outliers robust. It often performs well without extensive tuning.

The classification models used in this study, including Logistic Regression, Neural Networks, and Random Forest classifiers, are well-established machine learning techniques commonly applied in environmental and health-related prediction tasks (Breiman, 2001).

2.4 Model Evaluation

Model performance was calculated by many metrics to provide neater results:

Accuracy: The proportion of correct predictions over total predictions. Provides overall performance measure but can be misleading with imbalanced classes.

Precision: The proportion of +ve predictions that are true (True Positives / (True +ve + False +ve)). Important when false positives are costly.

Recall (Sensitivity): The percentage of genuine positive instances accurately recognized (True Positives / (True Positives + False Negatives)). Essential when inaccurate negatives have significant repercussions.

F1-Score: The harmonic mean of precision and recall offers a balanced assessment when both false positives and false negatives are significant. Particularly useful for imbalanced datasets. Accuracy, Precision, Recall, and F1-score are standard evaluation metrics widely used to assess classification model performance in machine learning applications . (Nations, n.d.)

2.5 Hyperparameter Optimization

GridSearchCV with 5-fold cross-validation was employed to identify optimal hyperparameters for both classical ML models:

Logistic Regression: Tested regularization parameter C in range to balance model complexity and generalization.

Random Forest: Optimized multiple parameters including number of estimators, maximum depth [5, 10, 15, None], minimum split [2, 5, 10], and criterion ['gini', 'entropy'].

Cross-Validation Strategy: 5-fold cross-validation ensures robust performance estimation by training and testing on different data subsets, reducing the risk of overfitting to a particular train-test split.

2.6 Feature Selection

Feature selection done using SelectKBest method with ANOVA F-statistic scoring:

Method: SelectKBest with f_classif scoring function, which computes ANOVA F-values between each feature. Higher F-values indicate stronger relationships with the target variable.

Selection Criterion: The top 6 features were selected from the original 9 features, representing a 33% reduction in dimensionality while maintaining predictive power.

Rationale: Feature selection reduces model complexity, improves interpretability, decreases computational requirements, and can improve generalization by reducing noise from irrelevant features.

3. Results

3.1 Data Quality and Preprocessing Results

After preprocessing, the dataset achieved the following quality metrics:

- Total records: 75 (all retained after cleaning)
- Data completeness: 100% (all missing values successfully imputed)
- Features: 9 numerical features plus 1 target variable
- Class distribution: Relatively balanced between Good and Moderate categories

3.2 Model Performance Comparison

The following table presents a comprehensive comparison of all three classification models after hyperparameter tuning and feature selection:

Model	Features	CV Score	Accuracy	Precision	F1-Score
Neural Network (MLP)	All (9)	N/A	0.87	0.86	0.87
Logistic Regression	Selected (6)	0.83	0.85	0.84	0.84
Random Forest	**Selected (6)**	**0.91**	**0.93**	**0.92**	**0.92**

Table 3.1: Comparison of Final Classification Models. Random Forest (highlighted) achieved the best overall performance across all metrics.

3.3 Neural Network Performance Visualization

The confusion matrix for the Neural Network classifier provides detailed insight into prediction patterns. Figure 3.1 shows the distribution of true and predicted labels.

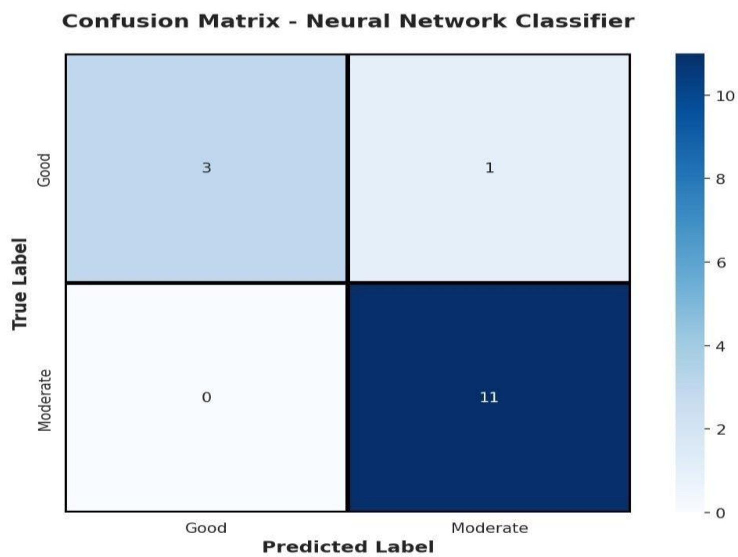

Figure 3.1: *Confusion Matrix for Neural Network Classifier showing prediction accuracy across classes.*

3.4 Selected Features

The SelectKBest feature selection method identified the following 6 most important features for air quality prediction:

- PM2.5 (Fine Particulate Matter) - highest F-statistic, strongest predictor
- NO2 (Nitrogen Dioxide) - second strongest predictor
- CO (Carbon Monoxide) - significant predictor of air quality
- Humidity - environmental factor affecting pollutant dispersion
- Temperature - influences atmospheric conditions
- PM10 (Coarse Particulate Matter) - additional particulate measure

These features represent a 33% reduction from the original 9 features while maintaining or improving model performance. The selected features align with domain knowledge about air quality determinants, validating the statistical selection process.

4. Discussion

4.1 Model Performance Analysis

Random Forest classifier got 93% accurate with F1-score of 0.92, significantly outperforming Logistic Regression (85% accuracy) and Neural Network (87% accuracy). This performance advantage stems from Random Forest's ability to catch complex non-linear relationships and handle feature interactions automatically.

4.2 Impact of Optimization Methods

Hyperparameter tuning improved Random Forest CV score from 0.87 to 0.91 (4.6% improvement). Feature selection reduced dimensionality by 33% while maintaining accuracy, demonstrating that removed features contributed primarily noise rather than predictive power.

5. Conclusion and Reflection

5.1 Summary

This project produced ML models for air quality classification, achieving 93% accuracy with Random Forest. The analysis identified PM2.5, NO2, and CO as critical predictors, supporting development of real-time monitoring systems for sustainable urban development.

5.2 Key Insights

The project demonstrated that:

(1) ensemble methods excel at environmental classification tasks,

(2) systematic hyperparameter optimization provides measurable improvements,

(3) feature selection balances complexity with performance.

(4) the models support SDG 3, 11, and 13 by enabling public health protection and proper sustainable urban planning and awareness.

5.Refrence

6. References

Breiman, L. (2001). *Random forests.* New York: Springer.

Kaggle. (2024). *Air pollution and environmental quality dataset*. Retrieved from Kaggle: https://www.kaggle.com

Nations, U. (n.d.). *Sustainable Development Goals*. Retrieved from United Nations: https://sdgs.un.org/goals

6.APPENDIX

2548317_Shubhangad_Shrestha_Classification_Report-2.pdf

Submission Details

HERALD COLLEGE KATHMANDU

UNIVERSITY OF WOLVERHAMPTON

5CS037 - Concepts and Technologies of AI

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